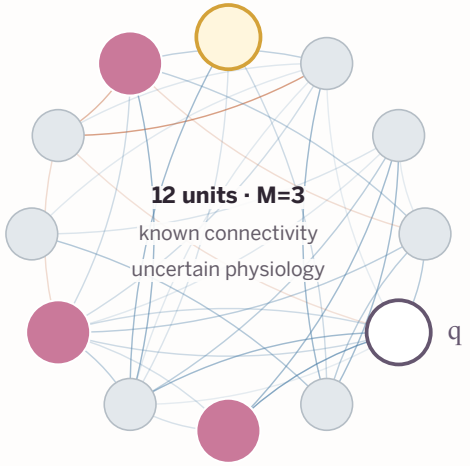
Predictive Risk and Recording Decisions

in Partially Observed Neural Networks

Nominal versus prior-averaged local risk for stimulus-response prediction



Blue: positive; orange: negative connections

Recorded Stimulus Readout

Integrated stimulus response

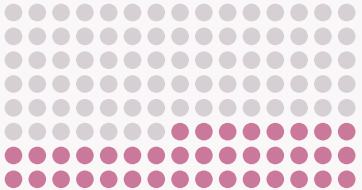
$$q = \int \Delta r(t)\, dt$$

$$\Delta r(t) = r_{\text{stim}}(t) - r_{\text{control}}(t)$$

1 Recording agreement

Nominal vs prior-averaged local risk
40 independent networks

68% 82 of 120 single-target designs are identical



65% 26 of 40 shared-target designs are identical



same recording set

2 Response error

Common MAP fit and budget
Same 40 networks

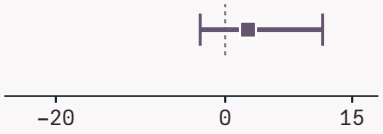
Single response

−3.7% [−18.5, +12.1]



Shared three-response

+2.7% [−3.0, +11.5]



Change vs nominal (%)

97.5% paired network-cluster CI

3 Risk forecasts

Fixed data and point prediction
40 new independent networks

$$S(v, e) = \log v + e^2 / v$$

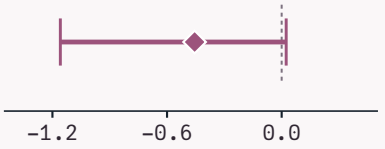
v: MSE forecast; e: response error

19 of 40 favour prior averaging; 21 favour nominal

One network supplies 64% of the summed score difference

MSE-forecast score difference

−0.46 [−1.16, +0.02]



Prior-averaged minus nominal

95% paired network-cluster CI

Identical recording sets and data yield identical predictions under the common MAP fit.
All primary comparison intervals include zero.

Exploratory 100-unit rank-two family (8 independent networks): four of eight 95% intervals exclude zero, all favouring prior averaging.
Intervals are unadjusted; one point estimate favours nominal.